

Classification of Indian Medicinal Plant Species Using Attention Module With Transfer Learning

Sivappriya.K

School of Artificial Intelligence

Amrita Vishwa Vidyapeetham,

Coimbatore, India

cb.en.p2dsc22005@cb.students.amrita.edu

ORCID: 0009-0009-7297-8458

Mithun Kumar Kar

School of Artificial Intelligence

Amrita Vishwa Vidyapeetham,

Coimbatore, India

k_mithunkumar@cb.amrita.edu

ORCID: 0000-0002-6326-4068

Abstract—Accurate plant species classification is an important task in the automatic diagnosis of medicinal plants because it increases the diagnostic capacity of subsequent processing. The key predictable characteristics of a medicinal plants are their form, color, and texture of its leaves. This publication presents an novel approach to classify medicinal plant species using attention-based deep convolutional neural networks. The proposed model has a convolution block attention module (CBAM) that achieved multi-class classification with 92.10% accuracy, 92.0% precision, 92.0% recall, and 92.0% F1-Score respectively. During the preprocessing stage, images are separated from their backgrounds using color channel processing and morphological operations. The proposed classification method is compared with state of art methods. Also the classification is done thorough transfer learning (Dense-Net 121 with accuracy of 99.66%, Vision transformer with accuracy of 97.0%, and Swin transformer with accuracy of 92.44%). The plant species are identified using 5945 images representing 40 different plant species from the DIMPSAR (Dataset for Indian medicinal plant species analysis and recognition) database, resulting in forty classes. In addition, research was conducted to investigate the influence of segmentation. The proposed model can be utilized to precisely categorize plants in computer-aided systems and to investigate therapeutic difficulties.

Index Terms—Indian Medicinal plants, CNN, CBAM, Attention Networks, DenseNet-121, Vision transformer, Swin transformer.

I. INTRODUCTION

Herbal base plant species used in ancient medicine have been extensively researched due to its nutritionally active compounds and therapeutic characteristics. Medicinal plants are valued for their active constituents and unique medicinal properties [1]. Plants are a crucial part of life on Earth. It provides us with nutrition, medicines, protection, and a healthy breathable environment [2]. The nature of therapeutic plants distinguishes them from chemical drugs, as does the fact that they have fewer adverse reactions. The leaf is regarded as the most critical component of the plants morphology which is utilized for identification. They are also recognized as a fundamental and necessary component of identification process of plants because they are near easily accessible throughout the year [3]. In computer aided diagnosis of medicinal plants, the plant identification has relied on a range of leaf characteristics, including its form, color, and texture. Leaf shape is an important characteristic since it distinguishes different

types of plants. [6]. The plant leaf has been extensively used in plant science for analyzing health. Leaf shape and texture are crucial for identifying the types of plants [5]. Manual identifying plant species has become arduous and time-consuming. Understanding plant shapes, petal and leaf types, automatic diagnosis can be possible through machine learning algorithms or deep learning algorithms [7]. The deep learning approaches extract valuable features from current data using deep layers of convolution neural network(CNN). Their self feature extracting capability made them robust and accurate for learning algorithms [8].

Automatic plant identification uses machine learning techniques for classification of gathered plant parts into various botanical plant species. Most hurdle occurs from inter-class similarities within a plant species and strong intra-class variations in the occlusion, background, posture, color, and illumination. Figure 1 represents image examples with inter-class similarities and intra-class variations.

The present work uses convolutional neural network (CNN) based attention model to classify indian medicinal plants into 40 classifications. Also transfer learning approaches are adopted to increase the classification accuracy. In this work, both methodologies are used to classify 40 different categories of medicinal plants. The DIMPSAR (Dataset for Indian medicinal plant species analysis and recognition) database [14] is utilized to classify diverse medicinal plants. In the image pre-processing stage, the plant leaf pictures undergo segmentation from the background using color-based image segmentation. Different CNN models are used with a transfer learning strategy to classify plant leaves [11]. The plant categorization model is tested on 595 leaf images from various classes. To verify the efficiency of the proposed methodology, different performance measures (including accuracy, specificity, and sensitivity) are used and tabulated. The proposed technique can easily and accurately aid to classification of therapeutic plants. The current study aimed to achieve robust and accurate identification of medicinal plants using computer vision processing algorithms rather than cost increasing manpower and time-consuming methods.

The succeeding parts of the manuscript is organized as follows: Section II describes the related works, Section III



Fig. 1. Images exhibiting inter-class similarities and intra-class variations. First row represents pepper plant. Second row represents betel plant.

outlines the plant pre-processing, segmentation, data augmentation, proposed architecture application, and transfer learning applications. Section IV outlines the suggested technique for plant image classification through deep learning and impact of segmentation. At last, discussion and conclusion are presented in Section V and VI respectively.

II. RELATED WORKS

Rahim Azadnia et al. [1] developed a comprehensive automated image processing technique to quickly identify ayurvedic medicinal plant category under verified and strictly controlled lightning conditions. They have used Artificial neural network (ANN) algorithm to extract shape, colour and texture information from the obtained images of medicinal plants. ANN were used to categorize diverse groupings of medicinal plants, and the most efficient neural network classifier were chosen based on accuracy, correlation and error. The efficient features were fed into the model, resulted in the optimal classifier model. The model's accuracy was 100%, with a correlation coefficient of 1.00 and a mean square error of 2.35×10^{-12} .

C. R. Alimboyong et al. [7] used a dataset of 4,234 plant images. Authors trained a CNN model and augmented the dataset to identify 12 plant species utilizing image transforms such as resize, rotate, flipping, histogram equalization and scaling. The trained CNN model achieved accuracy categorization of 99.74% during validation and 99.69% during testing, with sensitivity of 99.0%.

B.R.Pushpa et al. [11] developed an unbiased lightweight deep CNN (Ayur-PlantNet) for categorization forty ayurvedic plant species, with 92.27% Accuracy. Based on a comparison of pre-trained models Resnet-34, Resnet-50, VGG16, MobileNet-V3 Large, EfficientNetwork-B4, and Densenet-121.

R. Akter et al. [12] suggested a convolution neural network with dropout, gaussian noise, and batch normalization,

achieved 71.3% accuracy, 66.4% precision, recall 67.6%, and 66.0% F1-score for 10 class classification.

M.R. Dileep et al. [13] presented a dataset from kerala that included forty medicinal leaves. A deep neural network that was enhanced from alexnet were used to extract features efficiently. Finally, the classification was done with softmax activation and support vector machine classifiers.

Singh et al. [21] proposed a new method for distinguishing plant leaves utilizing color segmentation and the SLIC algorithm. The Inception V3 Model were used, with some modifications. Tan et al. [22] developed a method to categorize leaf images using characteristics extracted from three CNN models: transfer learning with AlexNet, fine-tuned AlexNet, and D-Leaf. Five machine learning approaches were used to identify features: SVM, ANN, KNN, NB, and CNN. For benchmarking. The morphometric approach was used to produce morphological metrics from sobel segmented veins. The proposed leaf model achieved 94.88% testing accuracy, comparable to the tuned AlexNet models. Sahithya et al. [30] proposed a novel network for unhealthy leaf identification using SVM and artificial neural networks.

Anchitaalagammai et al. [24] developed a method to identify and categorize medicinal plant species. This method found and classified five different indian medicinal plant species. The dataset contained 58,280 pictures, with around 10,000 per species. Leaves were identified based on texture, shape, color, physiological, and morphological features. The dataset was trained with the CNN architecture, which produced a high accuracy of 96.6%.

Makanapura et al. [25] suggested a plant seedling categorization utilizing a dataset containing images of 12 different species: three plant species and nine weed species. They build the classification framework with three distinct architectures: ResNet50V2, MobileNetV2, and EfficientNetB0. They trained the models with transfer learning and then compared their performance on a 833 images in testing dataset. They evaluated the deep learning models and found that EfficientNetB0 surpasses them all, having 96.26% of average F1-score and 96.52% of accuracy.

Thiru Siddharth et al. [26] applied transfer learning and a pre-trained classifier VGG-19 to access color images of swedish leaves and identified plant species. The four primary classification methods used were image pre-processing, image augmentation, feature extraction, and recognition, all of which were carried out during the model evaluation. The model was linked to elements of the Swedish leaf dataset, which included fifteen tree classes, and aided in predicting the appropriate class of an unknown plant with an accuracy of 99.70%.

III. METHODOLOGY

This manuscript proposes a novel architecture for medicinal plant species classification using attention based CNN model and transfer learning. Initially the plant images are segmented from the background using colour based image segmentation which is shown in Figure 2.

Figure 3 represents the working flow of the proposed work for image segmentation and classification of indian medicinal plant species.

A. Preprocessing

Pre-processing is the first step for classification task to enhance the model performance. Color channel thresholding and morphological approaches are used to shrink and segment the color leaf images from their backgrounds. Horizontal and vertical flips, as well as rotations, are implemented for augmenting the training images. To keep the aspect ratio, the image resolutions are kept at 224×224 pixels.

B. Methodology for Segmentation

Image segmentation is an initial and crucial stage in image processing that divides a sample images into separate segments and processes only the most desirable parts of the images. Pre-processing requires segmenting plants from background to avoid difficulties during dataset categorization.

Furthermore, this allows the model to acquire best features without needing to learn unrelated information. we have used color-based segmentation.

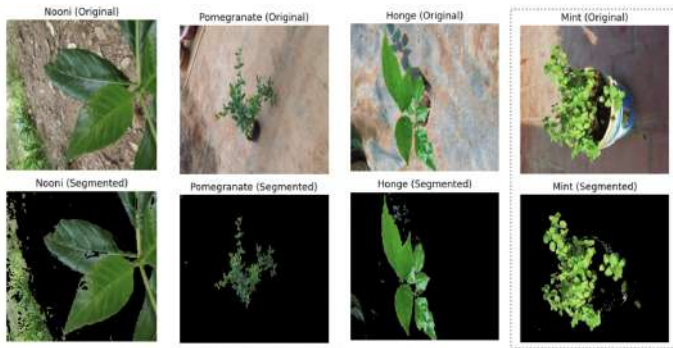


Fig. 2. Color based image segmentation of Indian medicinal plant species

The segmentation model recovers the plant region from the complicated background of low-resolution using color channel thresholding and morphological techniques. Images are converted from RGB color space to HSV color format. The HSV color space is preferred for color segmentation tasks because it distinguishes between intensity and color data. Table I displays the range of the HSV color space.

TABLE I
BOUNDS (LOWER AND UPPER) FOR HUE, SATURATION, AND VALUE

Parameter	Lower Bounds	Upper Bounds
Hue	30 (greenish-yellow)	90 (greenish-blue)
Saturation	50 (moderately saturated)	255 (fully saturated)
Value	50 (moderate brightness)	255 (maximum brightness)

The HSV color space is thresholded using a specific range of green channel. The image is transformed to binary, with pixels in the specified range assigned to white (255) and those outside the range set to black (0). This approach aids in

the separation of green-colored areas in images. The original image is combined with the thresholded image using a bitwise AND operation. This technique keeps just the portions of the original image when the matching pixel in the thresholded image is white (i.e., within the green color spectrum).

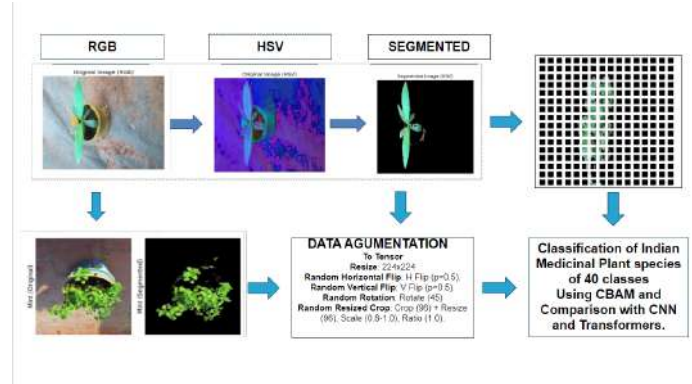


Fig. 3. Proposed methodology for image segmentation and classification.

C. Data Augmentation

The input images are re-sized to 224×224 to maintain the similarity between the images of different databases. To enhance the model performance the input training data is augmented using horizontal flipping, vertical flipping and rotation. The indian medicinal plant species from DIMPSAR dataset has been used for classifying different plant species using the proposed model.

D. Network Architecture for Proposed Classification

The proposed model uses numerous convolutional layers and attention mechanism to classify indian medicinal plant species. The attention mechanism emphasizes the attentive features by weighting the convolutional kernels. They allow for the modeling of dependencies regardless of their position in the input or output sequences.

The primary purpose of a CNN model is to extract local features from a layer by using hierarchy of self-learned attributes from earlier layers of the network. The main function of this model is to extract relevant information or features using attention mechanism within the CNN model. Hence channel and spatial attention both are required to extract intermediate features. [17]

The CBAM-Convolution block attention mechanism is integrated with two unique attention mechanisms to make use of critical channel and spatial axes properties. This increases the model's performance to extract feature information. The CBAM-integrated network enhances the capacity to effectively emphasizes target regions and extract features from that data. Furthermore, incorporating CBAM into the network increases both the identification of local as well as global features through attention mechanism.

SAM (Spatial Attention Module): This module consists of average pooling (AP) and maximum pooling (MP) blocks,

followed by convolution operation [28]. Mathematically it can be represented by Eq. 1.

$$M_s(F) = \sigma_{(f^{5 \times 5})} ([AP(F), MP(F)]) = \sigma_{(f^{5 \times 5})} ([F_{avg}^S, F_{max}^S]) \quad (1)$$

where, $M_s(F)$ represents a spatial attention map, $\sigma_{f^{5 \times 5}}$ represents an activation function (e.g., sigmoid) applied element-wise to the 5×5 convolutional layer.

For input feature size of the size $H \times W \times C$, MP and AP are processed along channel dimension to yield channel size of $H \times W \times 1$. The channels are merged followed by sigmoid activation function with range 0 to 1 with 0.5 threshold is used to calculate the weight coefficient M_s . Finally, the input feature is F multiplied by M_s to get new characteristics. Figure 4 shows the diagram of spatial attention module.

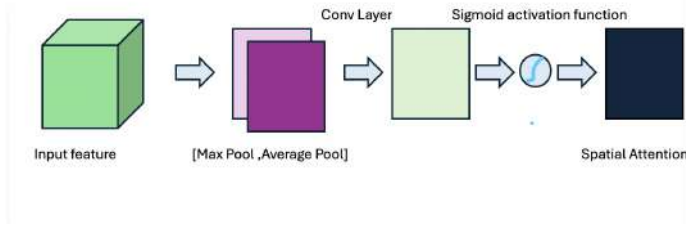


Fig. 4. Spatial Attention Module

CAM (channel attention module): In this module, attention mechanism is employed channel wise to capture relevant information. For input feature map F of the size $H \times W \times C$, global maximum pooling and average pooling is employed with $1 \times 1 \times C$ convolution kernels channel wise and merged to create attention scores using sigmoid activation function [28]. Figure 5 shows a diagram of CAM. Mathematically it can be represented by Eq. 3.

$$M_c(F) = \sigma(\text{MLP}(\text{AP}(F)) + \sigma(\text{MLP}(\text{MP}(F)))) \quad (2)$$

$$= \sigma(w_1(w_0(F_{avg}^c)) + w_1(w_0(F_{max}^c))) \quad (3)$$

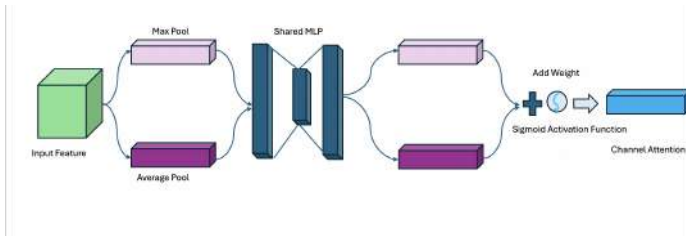


Fig. 5. Channel Attention Module

CBAM utilizes both SAM and CAM to maximize feature discrimination by multiplying attention weights to the specific convolutional layers. CBAM is included into the segmentation network, sandwiched between each convolution block. Figure 6 shows a diagram of CBAM.

The proposed architecture is consisted of a CNN model with attention blocks and a linear layer. The convolutional

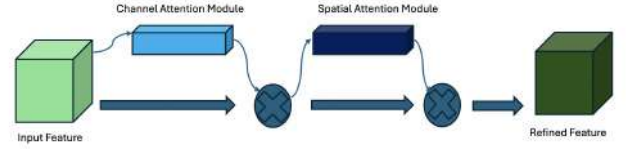


Fig. 6. Channel Block Attention Mechanism

block attention module (CBAM) for feature recalibration combines channel and spatial attention methods. CBAM improves feature representation by automatically recalibrating feature maps using channel and spatial attention techniques. Linear layer is fully connected with ReLU activation followed by dropout and is designed for classification with 40 classes. The CNN with CBAM has 393,510 trainable parameters. The linear layer contains 18,895,400 trainable parameters. Memory and computational efficiency comprises the forward/backward pass size for the CNN with CBAM is expected to be 183.75 MB, but for the linear layer, it is only 0.01 MB. Table II shows the information about layers of proposed architecture. Figure 7 shows the proposed CBAM network architecture details for classification. Figure 9 depicts the confusion matrix for categorization of test dataset using the suggested CBAM architecture. Table VIII and Table IX shows performance matrices using CBAM on dataset without and with segmentation.

TABLE II
LAYER INFORMATION WITH CBAM

Layer Type	Number of Layers
Convolutional Layers	3
Pooling Layers	3
Fully Connected Layers	2
Batch Normalization Layers	3
Activation Layers	2
Adaptive Pooling Layers	3
Channel Attention Layers	3
Spatial Attention Layers	3
Sigmoid Layers	4
CBAM Layers	3
Total Layers	28

E. Transfer Learning

Transformer-based models are used to categorize indian medicinal plant species via transfer learning. The Vision transformer (ViT) utilizes self-attention mechanism to compute input and output representations [20] and long range dependencies to capture global relationship between features [18]. To process 2D images, 2D Patches are extracted from the image and regenerated to one dimensional arrays and introduced into the positional encoding system for completion of patch embedding for next layer. The positional encoder ensures sequential positional occurrence of the patches one to each other. Multi-head attention layers yield linear combination of the attention heads [23]. Refer to equations 4–8 for mathematical formulation about vision transformer architecture. [23].

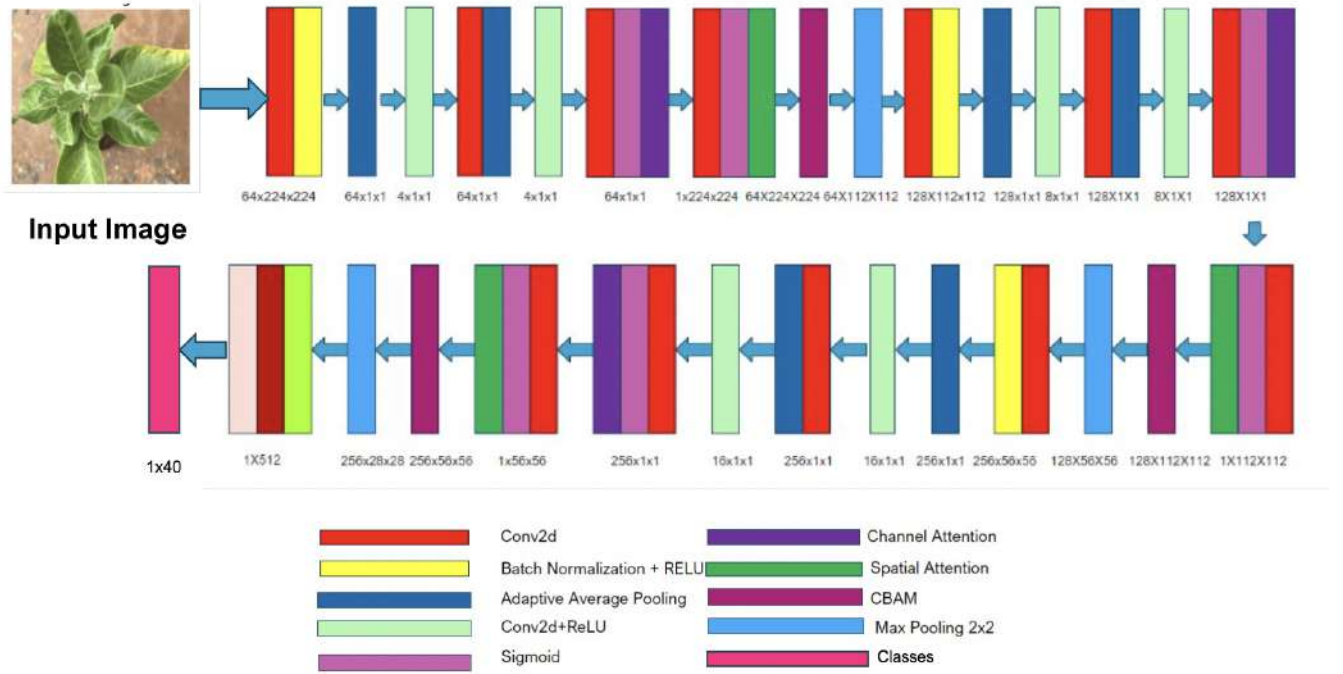


Fig. 7. Proposed CBAM network architecture for classification of Indian Medicinal plant Species

Figure 8 shows the vision transformer network architecture for classification of plant species.

Table IV shows the architecture implementation of vision transformer for classification of Indian Medicinal Plants.

1. Patch embeddings:

$$\text{Patch embeddings} = X \cdot W^p + B^p \quad (4)$$

Where:

- X is the input image.
- W^p represents the learnable weight matrix for patch embeddings.
- B^p represents the learnable bias for patch embeddings.

2. Positional Encodings:

$$\text{Positional Encodings (PE)} = \begin{cases} \sin(\frac{i}{10000^{2j/d}}) & \text{if } j \text{ is even} \\ \cos(\frac{i}{10000^{2j/d}}) & \text{if } j \text{ is odd} \end{cases} \quad (5)$$

Where:

- i represents positional variable.
- j represents dimension.
- d represents the dimensionality of the embeddings.

3. Self-Attention Mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V \quad (6)$$

- The matrices Q , K , and V represent the query, key, and value, respectively.
- d_k represents the key's dimension.

4. Multi-Head Attention:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (7)$$

Where:

- $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$ for $i = 1, \dots, h$. The weight matrices for linear projections are W_i^Q , W_i^K , and W_i^V .
- W^O represents the output's learnable weight matrix.

5. Feedforward Network:

$$\text{FFN}(x) = \text{ReLU}(xW_1 + B_1)W_2 + B_2 \quad (8)$$

Where:

- W_1 and W_2 are the learnable weight matrices for the first and second linear layers, respectively.
- B_1 and B_2 are the learnable biases.

Furthermore, Swin transformer [19] (hierarchical vision transformer with shifted windows) is utilised. The sliding approach enhances efficiency by minimizing computation of self-attention mechanism to distinct local windows while yet permitting cross-window connections. Figure 10 shows the swin transformer network architecture for classification of plant species. It generates hierarchical feature maps by combining picture patches in deeper layers, and its calculation complexity is proportional to input image size because self-attention is computed only within each local window. [16]. Table III shows the swin transformer model performance and metrics with segmentation.

TABLE III
SWIN TRANSFORMER MODEL PERFORMANCE AND METRICS WITH SEGMENTATION

Model Performance		
Hyperparameter	Value	
Loss Function	Cross entropy loss	
Optimizer	Stochastic gradient descent	
Learning rate	0.01	
Performance metrics		
Precision	Recall	F1-score
93.0	93.0	93.0
Test accuracy	92.44	

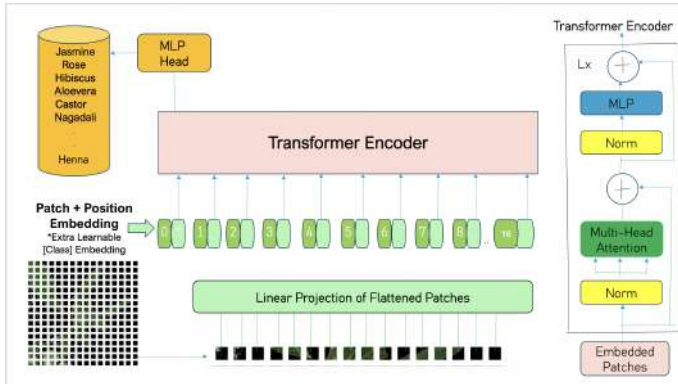


Fig. 8. Vision Transformers Architecture For Indian Medicinal Plant Species

IV. EXPERIMENTAL RESULTS

A. Dataset Description

This work is implemented using DIMPSAR dataset [14] (a dataset for analyzing and recognizing Indian medicinal plant species). The plant species collected include herbs, shrubs, and trees. It is developed by acquiring images from botanical gardens and medicinal plant farms in south karnataka and kerala. Images are obtained using cellphones having resolutions ranging from 2MP to 16MP. The dataset includes 40 plant species, 5200 rawest images, and 5945 post augmented image. The criteria followed to divide the sample of datasets for training, test, and validation are 80-10-10 split which include 4756 training images, 595 test images and 594 validation

TABLE IV
MODEL SUMMARY OF VISION TRANSFORMER

Component	Parameters	Description
ViT	Embeddings:	742,656 parameters
	Encoder:	85,054,464 parameters
	LayerNorm:	1536 parameters
Classifier	Linear	30,760 parameters
Image and Patch Details		
Image Size	224 × 224 pixels	
Patch Size	16 × 16 pixels	
Head	12	
MLP size	3072	

TABLE V
COMMON HYPER-PARAMETERS FOR TRAINING WITH AND WITHOUT COLOUR BASED SEGMENTATION

Hyper-parameter	Value
Batch size	32
Loss function	Cross Entropy
Optimizer	Adam
Learning rate	0.001
Epochs	50

images.

B. Results

Color segmentation is used to segment plant species, which are then classified using a convolution neural net with attention networks (CBAM). The results were compared to state-of-the-art pretrained vision transformers, swin transformers, data efficient image transformers, and pretrained convolution neural networks. Table V represents the common hyper parameters detail used for training with and without colour based segmentation using different models. Table VI and VII show the evaluation metrics of various deep learning models on medicinal plant species dataset.

TABLE VI
PERFORMANCE METRICS OF VARIOUS MODELS WITH ACCURACY AND PRECISION ON INDIAN MEDICINAL PLANT DATASET

Model Name	With Segmentation		Without Segmentation	
	Accuracy	Precision	Accuracy	Precision
Vision Transformer	95.97	96.0	73.45	76.0
Dense Net 121	95.63	96.0	99.66	100
Data Efficient Image Transformer	69.08	71.0	88.40	90.0
EfficientNet-b0	94.62	95.0	98.49	99.0
Resnet-101	94.79	95.0	90.76	90.0
Inception V3	90.42	91.0	92.10	94.0
Xception-Net	95.46	96.0	99.50	99.0
Vgg-16	44.20	47.0	67.06	68.0
Mobile-Net	92.61	93.0	91.07	92.0
CoAtNet	77.98	82.0	83.19	85.0

TABLE VII
PERFORMANCE METRICS OF VARIOUS MODELS WITH RECALL AND F1-SCORE ON INDIAN MEDICINAL PLANT DATASET

Model Name	With Segmentation		Without Segmentation	
	Recall	F1-Score	Recall	F1-Score
Vision Transformer	96.0	96.0	73.20	72.0
Dense Net 121	95.0	96.0	100	100
Data Efficient Image Transformer	69.0	68.0	88.0	88.0
EfficientNet-b0	95.0	95.0	99.0	98.0
Resnet-101	95.0	95.0	89.0	89.0
Inception V3	90.0	90.0	92.0	92.0
Xception-Net	95.0	95.0	99.0	99.0
Vgg-16	46.0	44.0	68.0	66.0
Mobile-Net	93.0	93.0	92.0	92.0
CoAtNet	78.0	77.0	84.0	83.0

C. Evaluation Metrics

The evaluation metrics are used to evaluate the categorization outcomes of the test dataset. The different matrices used in the manuscript are accuracy (Acc), precision (Pre), recall, and F1-score. [29] Furthermore, a confusion matrix of test set findings is displayed to check and analyze the results.

TABLE VIII
PERFORMANCE METRICS USING CBAM ON INDIAN MEDICINAL PLANT
DATASET WITHOUT SEGMENTATION

Model	Optimizer	Epoch	Accuracy	F1-Score	Precision	Recall
CBAM	ADAM	50	87.56	87.0	88.0	87.0
CBAM	ADAM	100	91.60	91.0	92.0	92.0
CBAM	SGD	100	87.39	87.0	88.0	87.0
CBAM + DenseNet Blocks	ADAM	100	92.10	92.0	92.0	92.0
CBAM + DenseNet Blocks	ADAM	50 (3-fold CV)	87.0	87.0	89.0	86.0

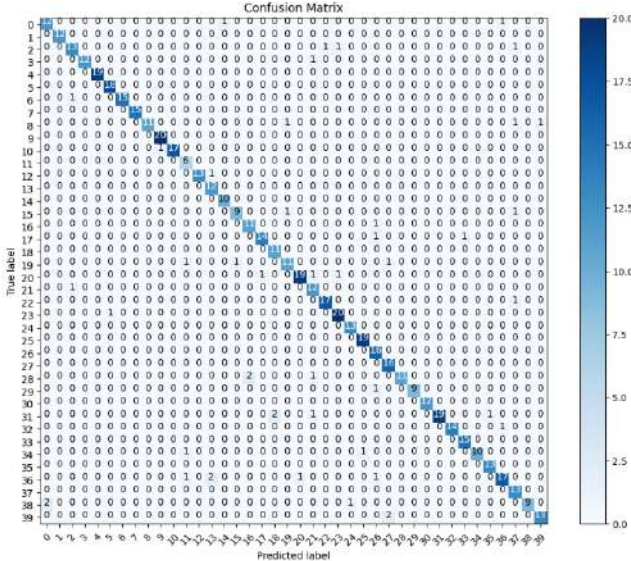


Fig. 9. Confusion Matrix For Indian Medicinal Plant Species Test dataset Using proposed CBAM Architecture

To completely assess the success of a technique, one needs examine both precision and recall. Unfortunately, striking the right balance between precision and recall is tough. Attempting to improve precision often reduces recall, and vice versa. [31] The F1-score is also known as the weighted harmonic mean of the diversity and coverage metrics. F1-score is considered to acquire its greatest value at one, indicating flawless precision and recall scores, and it is worst when a value is zero.

V. DISCUSSION

Vision Transformer and Dense Net 121 outperform the other pre-trained CNN models in terms of metrics accuracy and precision, however data efficient Image Transformer falls short. Models such as Vgg-16 are less accurate and precise, particularly without segmentation. The implementation of CoAtNet yielded positive results, showing its potential efficacy.

CBAM architecture, when used alone or with DenseNet blocks, the CBAM architecture performs better with enhanced

TABLE IX
PERFORMANCE METRICS FOR CBAM + DENSENET BLOCKS AND VISION
TRANSFORMER WITH SEGMENTATION

Model	Epochs	Accuracy (%)	F1-Score (%)	Precision (%)	Recall (%)
CBAM + DenseNet Blocks	100	87.90	88.0	89.0	88.0
Vision Transformer	100	97.0	97.0	97.0	97.0

accuracy, precision, F1-score, and recall. Similarly swin transformer produces good outcomes in terms of evaluation metrics especially in segmentation. The swin transformer yields high precision and recall during segmentation, demonstrating its effectiveness at segmenting medicinal plant species.

The selection of an optimizer, such as SGD or Adam, for Vision Transformer with color-based segmentation has a significant impact on training. In this comparison, the scenario using the SGD optimizer performed better (95.97%) than the scenario using the Adam optimizer (70.25%). This suggests that SGD may be more effective at optimizing model parameters for this specific job and dataset. Finally, using DenseNet-121 for image classification tasks resulted in a considerable performance difference with and without color-based segmentation. With segmentation, it achieved 95.63% accuracy, 96.0% precision, 95.0% recall, and a 96.0% F1-score. In contrast, without segmentation, it performed far better, with accuracy, precision, recall, and F1-Score of 99.66%, 100%, 100%, and 100% respectively.

VI. CONCLUSION

In this manuscript, a novel method for automatic classification of different types of medicinal plant species is proposed using attention based deep neural network (CBAM). The network utilizes spatial as well as channel attention in CNN classifier for accurate classification of 40 types of clinical plants using Indian medicinal plant dataset. Also transfer learning is incorporated for the classification task using vision transformer and swin transformer models. In pre-processing process the plant images undergo segmentation from background by color based segmentation method. The CBAM model achieved accuracy of 92.10% however Dense-Net 121 achieved accuracy of 99.66%, Vision transformer achieved accuracy of 97.0%, and Swin transformer achieved accuracy of 92.44%. for medicinal plant classification. This models used in the manuscript can utilized for fast and accurate classification of plants in CAD systems.

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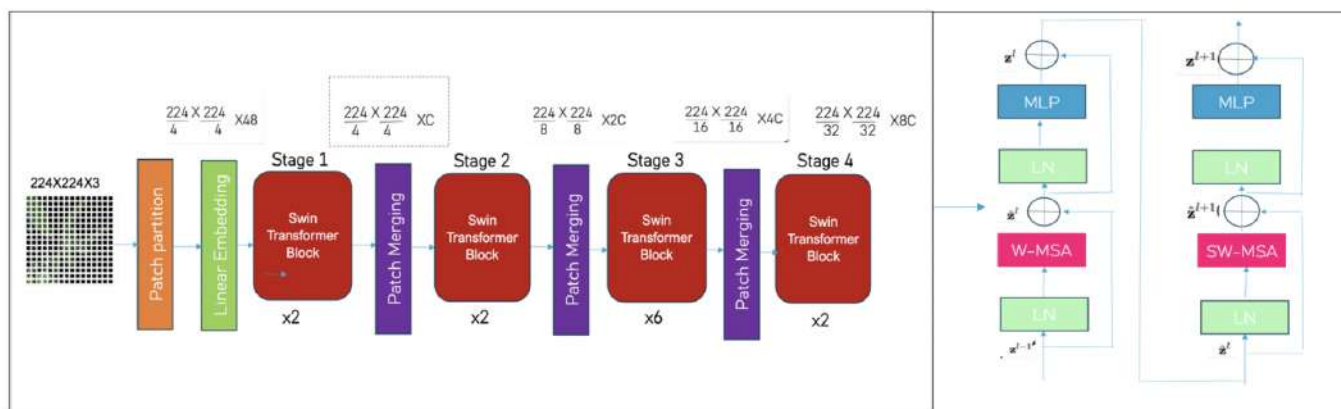


Fig. 10. Swin Network Architecture for classification of Indian Medicinal plant Species

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